Flag relations and incidence residues: complete manuscript costs

Human Compression project

The two fixed sources are Delaby–Leemans–Tranchida, *Geometries with trialities arising from linear spaces*, Algebraic Combinatorics 9 (2026), 379–402 (L009), and Leemans, *A generalization of a construction due to Van Nypelseer*, Beiträge Algebra Geom. 46 (2005), 561–574 (L092). Their complete PDFs have 25 and 14 physical pages; L009 includes its publisher identification cover. The accepted principal contracts contain 79 source records, with explicit corrections and exclusions. They do not assert recovery of every printed auxiliary claim.

The common theory and the two applications. The primary convention counts calibrated NFC characters without whitespace. Only exact running furniture is removed. Titles, abstracts, proofs, references and extractable table and diagram labels remain charged.

Complete document	Pages	Characters
Source L009	25	49891
Source L092	14	23308
Whole common theory U_G	12	30928
Triangle application G1	8	23544
Steiner application G2	22	61663
Whole program appendix	20	53018

The combined written comparison pays for the whole common theory once:

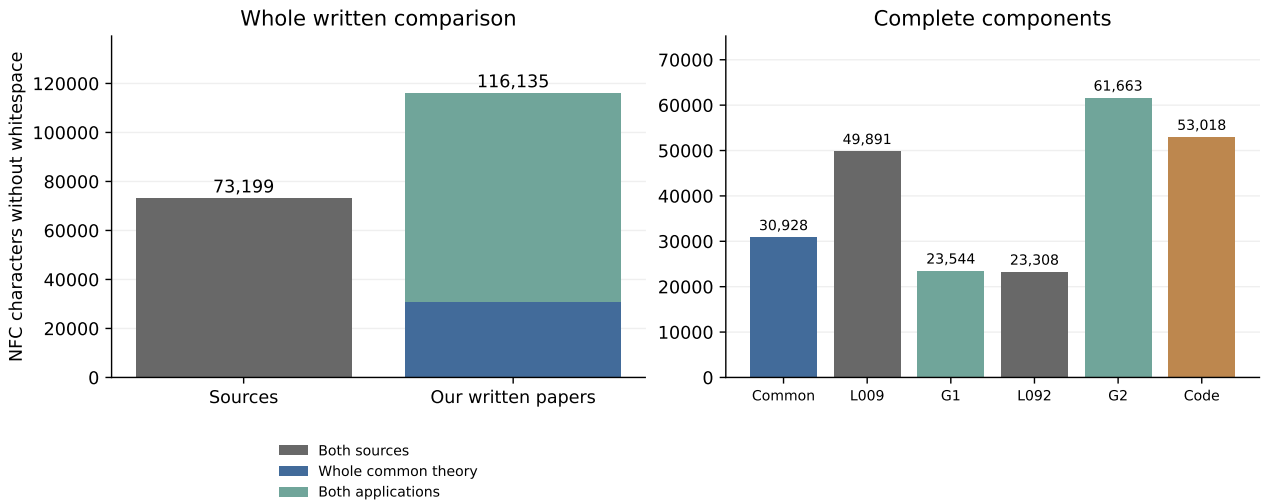
$$R_{\text{written}} = \frac{49891 + 23308}{30928 + 23544 + 61663} = \frac{73199}{116135} = 0.630.$$

After acquiring the whole common paper and all thirteen complete programs, the two source/application comparisons are

$$r_1 \mid (U_G, P_G) = \frac{49891}{23544} = 2.119, \qquad r_2 \mid (U_G, P_G) = \frac{23308}{61663} = 0.378.$$

The complete written replacement is longer than the two sources in this measure. G1 is shorter than L009; G2 is longer than L092 even after the common theory is acquired. Other topics pay their own common theories in full, even where they overlap.

Programs, formal evidence and retained inputs. Four programs supply finite mathematical inputs, two verify witnesses or bridge them to certificates, and seven validate separately proved arguments. All thirteen are retained in full. The printed appendix gives $73199/(116135 + 53018) = 0.433$. The thirteen raw files instead cost 50184, giving $73199/166319 = 0.440$. These are alternative costs, never simultaneous charges. Three drivers cost 14919 separately; six Lean files cost 19480 separately and contain forty previously checked component declarations. They do not formalize the entire source contracts. BDL, Kantor and other classical inputs remain attributed and their full proof costs unknown, as do historical programs, graphics and formula layout.



The left panel compares complete written totals. The right panel displays all common, source, application and printed-program costs separately. These lengths do not establish a recurring sawtooth pattern or minimum mathematical proof complexity.

Sensitivity. For calibrated selected text with NFC normalization:

Unit	Combined written	L009/G1	L092/G2
Whitespace words	0.728	2.513	0.499
Alphanumeric units	0.724	2.463	0.454
Nonspace characters	0.630	2.119	0.378
Characters, collapsed spaces	0.647	2.181	0.398
UTF-8 bytes, collapsed spaces	0.649	2.258	0.406

All five units are retained under NFC and NFKC for three extraction policies: untrimmed raw, selected raw and calibrated selected. The thirty coordinates supply 150 exact fractions: ninety combined and sixty conditional. Removing only L009's identification cover gives the separately counted article cost 49402. This changes the combined written factor to 0.626 and the L009/G1 factor to 2.098; L092/G2 is unchanged. The additional 120 exact fractions retain the printed and raw program alternatives too. The primary comparison continues to include that cover.

Source qualifications and extraction. L009 retains the determinant correction and its circle-list, polarity and ancillary-example limits. L092 retains the corrected deleted-line and subline cases, path convention and Figure 2 qualifications; named-geometry identifications and the claim without partial linearity remain excluded. The full source texts are counted with these declared principal-result contracts, not with an assertion that every historical statement is true.

The independent input audit checked all 101 physical pages, 180 PDF metric cells and 220 separate raw-file cells. It aligned 75,909 source characters with actual PDF text bytes and inspected the embedded glyph evidence. The corrected cache identifies actual font dictionaries and effective encodings: 701 occurrences had previously been mislabeled in the audit, but none of the metric cells changed. The 509 visible-glyph calibrations and all raw sensitivities remain explicit. The retained parsers are pypdf 6.13.0 and fontTools 4.63.0; extraction does not price complete visual information. Independent numerical review is required before publication.